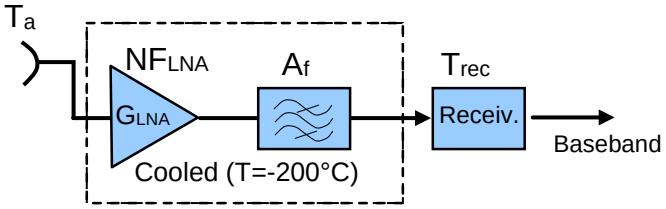
Exercise 4

A space probe is sent in orbit around Saturn to take high-quality pictures of the planet surface. The probe has a dish antenna (4m diameter, aperture efficiency e_a=0.6) and a transmitter with P_T=30 W at 10 GHz. The receiving station on Earth employs an antenna with 70m diameter (e_a=0.65), connected to the receiver schematically depicted in the following figure.

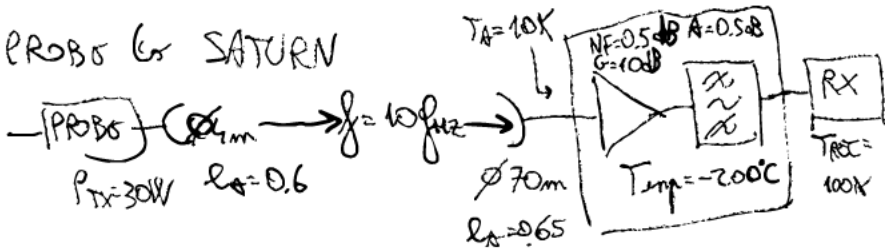


T_a=10°K
NF_{LNA}=0.5 dB
G_{LNA}=10 dB
A_f=0.5 dB
T_{rec}= 100 °K
K=$1.38 \cdot 10^{-23}$

1) Evaluate the gain of receiving and transmitting antennas
2) The distance Saturn-Earth is $1.43 \cdot 10^9$ Km. Compute the received power P_r at the antenna output
3) Evaluate the equivalent system temperature (T_{sys}) of the receiving system
4) Assuming the requested E_b/N_0=10 dB, evaluate the maximum data rate R allowed by the system (Eb/N0=10dB comes by selecting a BPSK modulation with BER=3.9e-6)
5) If the signal bandwidth *B* is 54kHz , what is the SNR_{sys}?
6) How long does it take to transmit 100Mbit of data?
7) If alpha (roll-off)=0.419, what modulation (i.e. BPSK, 4QAM, 16QAM...) is being used?

EXERCISE ANTENNA N.4 [90]

SPACE PROBE to SATURN



($K_B = 1.38 \cdot 10^{-23} \frac{W}{Hz K}$)

① Gain of ANTENNAS:

$Ø4m \rightarrow Area = \pi\left(\frac{Ø}{2}\right)^2 = 12.6m^2 \rightarrow A_e = \ell_A \cdot A = 7.56m^2$

$G_{TX} = \frac{A_e}{\lambda^2/4\pi} = 105'560 = 50.2dBi$

(NOT REQUIRED HPBW $= \pm 0.27°$)

$Ø70m \rightarrow Area = 3'848m^2 \rightarrow A_e = 2'302m^2$

$G_{RX} = 75.4dBi$

(HPBW $= \pm 0.015°$)

② Distance $1.43 \cdot 10^9$ Km → compute $P_{RX} = P_{TX} + G_{TX} - FSL + G_{RX} = -155.2$ dBW

(P_{TX} = 14.8 dBW)

$P_{RX} = -125.2 dBm$

$FSL = 20 \log \frac{4\pi d}{\lambda} = 295.5 dB$

③ T_{SYS}:

$T_{PHYS} = -200°C = 73K$ (Physical temp of filter, Tphys)

$T_{SYS} = T_A + T_{LNA} + \frac{T_{FLT}}{G_{LNA}} + \frac{T_{REC}}{G_{LNA} G_{FLT}}$

T0LNA=290K (always, by definition of NF)

$T_{LNA} = T_{0\,LNA}(NF - 1) = 35.4K$ $G_{LNA} = 10$

$T_{FLT} = 8.9K = T_{PHYS}\left(\frac{1}{G_{FLT}} - 1\right)$ $G_{FLT} = 0.8913$

$T_{SYS} =$ 57.5K

(4) $\frac{E_b}{N_0} = 10\,dB \Rightarrow$ MAXIMUM DATA RATE?

$P_{noise}^{eq} = K B T_{eq}$ $\quad P_{SIG\,RX} = -125.2\,dBm = 3.02 \cdot 10^{-13}\,mW$

$$SNR = \frac{P_S}{P_N} = \frac{E_b R_{bps}}{N_0 \cdot BW} \Rightarrow \frac{P_{SIG\,RX}}{K B T_{eq}} = \frac{E_b R_{bps}}{N_0 BW}$$

$$\Rightarrow R_{bps} = \frac{P_{SIG\,RX}}{K T_{eq} E_b/N_0} = 38.06\ Kbit/s$$

ACHIEVABLE IN MANY CHOICES of BW

(5) If $B = 54kHz$ what is SNR_{sys}?

$$SNR = \frac{E_b}{N_0} \frac{R_{bps}}{BW} = 7 = 8.45\,dB$$

(6) Time to transmit 100 Mbit data?

$$Time = \frac{100\,Mbit}{R_{bps}} = 2627s = 44minutes$$

(7) if $\alpha = 0.419$, HOW MANY Bits/symbol are used?

$BW = R_{sps} \cdot (1+\alpha)$

$R_{sps} = 38.06kSym/s$ (kBaud)



$R_{sps} = \frac{R_{bps}}{\log_2 M} \Rightarrow \log_2 M = 1 \Rightarrow M = 2 \Rightarrow$ BPSK